

Zelevtra x AWS ML Ascend Challenge

Advanced Ensemble Learning for Solar Panel Efficiency Prediction

Team Name: TensorV3

Members:

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Problem Statement

Objective:

Predict solar panel efficiency based on environmental and operational parameters

Dataset Characteristics

- **Target Variable:** Efficiency (continuous regression problem)
- **Features:** Environmental conditions (humidity, pressure, wind_speed) and operational parameters
- **Data Quality Issues:** Mixed data types, missing values, inconsistent formatting
- **Feature Engineering Opportunity:** Interaction terms and polynomial features

Key Preprocessing Challenges

- String-to-numeric conversion for meteorological data
- Handling mixed data types (categorical and numerical)
- Missing value imputation strategy
- Feature scaling and normalization

Advanced Preprocessing Pipeline

1. Data Cleaning & Type Conversion

Converted meteorological features (humidity, pressure, wind speed) from text to numeric format

Handled invalid values using coercion to ensure consistency

2. Power Transformation & Scaling

Yeo-Johnson Transformation: Normalize skewed distributions

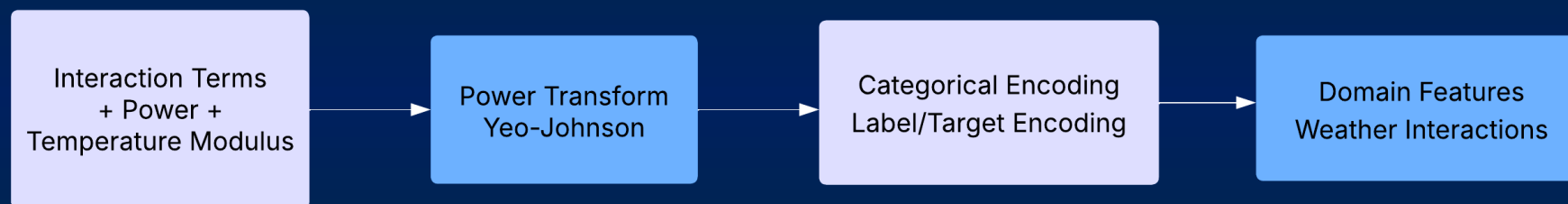
Robust Imputation: Mean strategy for missing values

Label Encoding: Categorical variables to numerical

Infinite Value Handling: Replace inf/-inf with NaN

Advanced Preprocessing Pipeline

3. Feature Engineering (Model-Specific)



4. Model-Specific Preprocessing

TabNet: Imputer Categorical embeddings with dimensions [3,3,3]

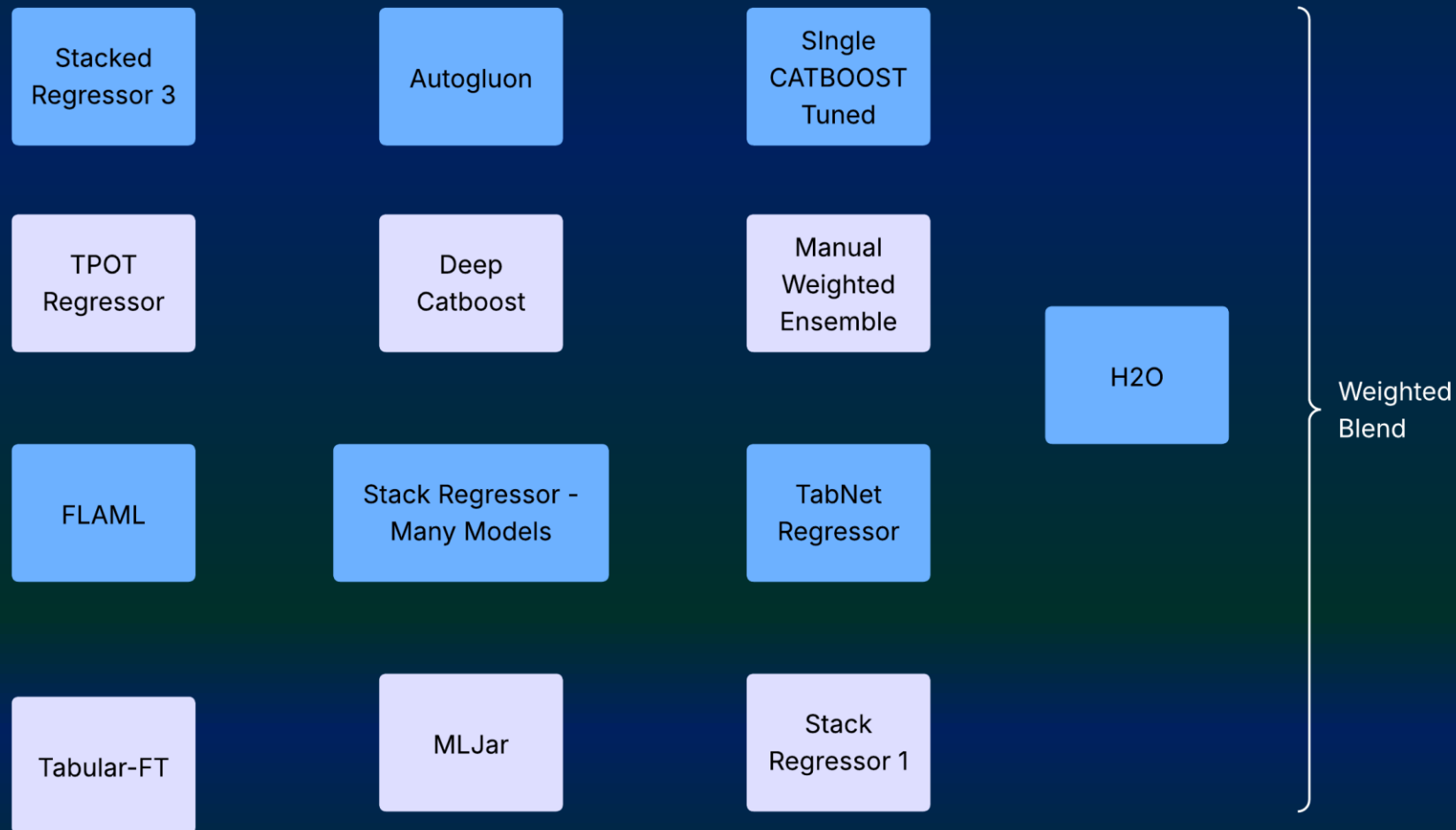
H2O/AutoGluon: Automated feature engineering

Tree Models ML: Standard scaling + polynomial features

Stack Regression Models: Standard scaling + polynomial features

Comprehensive Model Architecture

Multi-Level Ensemble Strategy



Comprehensive Model Architecture

1. **Stacked Regressor 3:** An Hyperparameter tuned ensemble regressor combining Catboost, XGBoost, RandomForest, LGBM, ET using stacking to leverage their complementary strengths.
2. **AutoGluon:** A powerful AutoML framework that automates model selection, training, and ensembling for tabular datasets.
3. **Single CatBoost Tuned:** A hyperparameter tuned high-performing gradient boosting model fine-tuned for optimal accuracy on structured and categorical data.
4. **TPOT Regressor:** An AutoML tool that uses genetic programming to discover the best-performing regression pipeline.
5. **Deep CatBoost:** An enhanced version of CatBoost leveraging deeper architectures.

Comprehensive Model Architecture

6. **Manual Weighted Ensemble:** A custom ensemble approach that linearly combines model outputs using manually chosen weights for better predictions - Catboost, XGBoost, RandomForest, LGBM, ET.
7. **FLAML:** A fast and lightweight AutoML library that efficiently searches for accurate models under resource constraints.
8. **Stack Regressor – Many Models:** A stacked regression model combining Gradient Boosting, Random Forest, XGBoost, LightGBM, and CatBoost is used to predict efficiency. Ridge regression acts as the meta-learner to blend base model outputs for improved accuracy.
9. **TabNet Regressor:** A deep learning model that uses sequential attention to perform interpretable and efficient tabular data learning. Hyperparameter tuned using optuna.

Comprehensive Model Architecture

10. H2O: A scalable and robust AutoML platform that automatically builds and blends multiple regression models.
11. Stacking Regressor – FLAML : Used FLAML as Meta-Model in stacking regressor
12. Stacking Regressor – FLAML Advanced: Tuned FLAML
13. Stacking Regressor 1 - An Hyperparameter tuned ensemble regressor combining Catboost, XGBoost, RandomForest, LGBM, ET using stacking to leverage their complementary strengths.
14. MLJar - The MLjar-AutoML is an Automated Machine Learning Python package that works with tabular data.
15. Tab FTTransformer – A hyperparameter tuned Transformer based Tabular ML Model

Advanced Multi-Level Ensemble Architecture

Manual Weighted Ensemble Breakdown

Model Name	Individual LB Score	Weight
H2O	89.89176	0.65
Manual Ensemble Weighted	89.86597	0.50
TabNet	89.88160	0.40
Stack Regressor 1	89.87982	0.40
AutoGluon	89.84440	0.30
Deep CatBoost	89.84000	0.35
MLJAR	89.88173	0.10

Advanced Multi-Level Ensemble Architecture

Manual Weighted Ensemble Breakdown

Model Name	Individual LB Score	Weight
Stack Regressor – Many Models	89.85290	-0.20
Stack Regressor 3	89.87988	-0.10
TPOT	89.83612	-0.20
FLAML	89.84897	-0.10
Tabular-FTTransformer	89.76028	-0.20
Single CatBoost (Tuned)	89.84074	-0.35
Stacking Regressor – FLAML Advanced	89.85412	0.1
Stacking Regressor – FLAML	89.84103	-0.1

Model Diversity and Ensemble Strategy

- Ensemble Innovation: Negative Weighting
- Diversity Enhancement: Negative weights prevent overfitting
- Bias Correction: Counteract systematic errors
- Performance Boost: Improved generalization through contrast
- Risk Management: Reduced ensemble variance

